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# Targeting seasonal climate forecasts for integration into household level decisions: the case of smallholder farmers in Lesotho

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Seasonal climate forecasts have been promoted as a means to increase the resilience of marginal groups in Africa. The manifestations of this are still to be seen. This paper argues that successful dissemination and adoption of the forecast requires an in-depth profile of the characteristics and needs of user groups. The case study of a mountainous village in southern Lesotho is used to highlight the decisions which one group of marginal users – smallholder farmers – might make in response to the forecast. A participatory role-play exercise explores what information households presently receive and how new climate forecast information could be integrated into seasonal decision making. Results show that there are a number of low-input options available to these farmers for responding to the forecast. Adoption, however, is going to require repeated exposure to the forecast in conjunction with forecast development that is suited to users' needs. The case study is linked back to the larger scale by suggesting paths that seasonal climate forecast development could take if it is to contribute to improving livelihood sustainability among marginal groups.

KEY WORDS: Lesotho, seasonal climate forecasts, participatory role-play methods, smallholder farmers, climate adaptation, rural livelihoods

## Introduction

The ability to adapt to a highly variable climate is one of the greatest challenges facing smallholder farmers in southern Africa because of their dependence on natural resources and agriculture (Scoones 1997; Jones *et al.* 2000; Vogel 2000). Seasonal climate forecasts, which are probabilistic predictions of how much rain is expected during the season, based on the principle that the ocean forces atmospheric behaviour (Washington and Downing 1999), predict some of this climatic variability. Farmers could potentially use forecasts to plan management strategies and resource allocation to reflect anticipated climate. These changes could be used as a means of household adaptation to climate variability at the seasonal time scale. If households could learn to adapt to uncertain environments by using forecast information to decrease the negative impacts of climate variability and capture the benefits of climate variability, pathways of increased livelihood security could be supported. Despite this potential, there are many constraints to using seasonal forecasts and few examples of how smallholder farmers have or have not adopted the forecast (Blench 1999). The implication is that it is difficult to know how forecasts could be used and what the needs and wants of this user community might be. For this reason, an exploratory method of engaging smallholder farmers in the reality of seasonal forecast applications is necessary.

During the past few decades climatologists have improved their ability to predict the seasonal climate (Cane *et al.* 1986; Palmer and Anderson 1994; Martin *et al.* 2000; Murphy *et al.* 2001).

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These seasonal climate forecasts predict temperature and rainfall. Rainfall predictions provide probabilities of the total amount of rainfall over a given time period (usually one to six months) being above, below or within the normal range based on past rainfall averages. The forecasts rely on the ocean–atmosphere inter-dependency and so sea surface temperatures provide an indication of future atmospheric perturbations (Washington and Downing 1999). This predictability is caused largely by tropical atmospheric circulation and is dependent on the El Niño/Southern Oscillation (ENSO)-related anomalies (Mason *et al.* 1996).

The information that seasonal forecasts provide can be used in many sectors. Research undertaken in southern Africa has shown that climate variability impacts on many aspects of the economy (Jury 2002). This paper is concerned primarily with rural livelihoods and the related agricultural practices. Forecasts could aid agricultural planning by enabling decisions involving climatic risk to be better informed (Hammer 2000). For example, forecasts could help farmers decide what type of crops to plant and when, what precautionary measures to take, whether to diversify or not, and in so doing could avoid undue risks (Sivakumar *et al.* 2000). If this is the case, then the importance of forecasts to marginal groups involved in agriculture should be a research priority. In regions such as sub-Saharan Africa, where the climate is highly variable and 20% of the gross domestic product (GDP) and more than two-thirds of the labour force are accounted for by agriculture (World Bank 1998), the forecast could play an important role. Not only does agriculture contribute to food security but also to employment and other aspects of livelihoods (Maxwell 2001), and so the impact of the forecast could be multi-dimensional.

The idea that predictive science can simplify decision-making processes by creating a clearer picture of the future is appealing in principle, but problematic in practice. The application of forecasts is hindered by a number of constraints that include issues related to temporal and spatial scale, the probabilistic nature of the forecast and ways of integrating the forecast into decisions (Nicholls 1999 2000; Stern and Easterling 1999; Broad 2000; Vogel 2000; Agrawala *et al.* 2001; Patt and Gwata 2002). Marginal groups do not always have the means to cope with these constraints and so using the forecast could make them more vulnerable. It is therefore necessary to consider what these constraints are at the same time as taking into account smallholder farmers' perception of the forecast and how they deal with uncertain information.

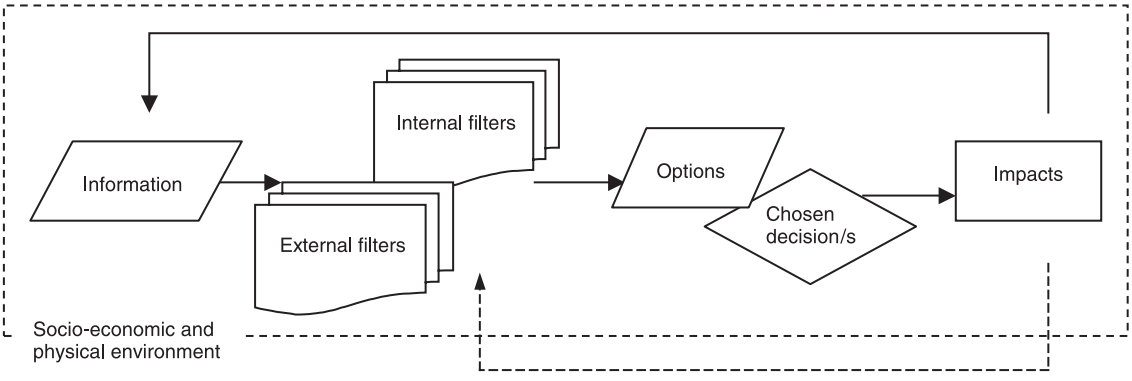
Although agriculture does not constitute a high proportion of national GDP in southern Africa, it is a sector that affects many people's livelihoods and so is a key area of seasonal forecast applications research. Research has shown that commercial farmers are interested in the seasonal forecast and are starting to use the information (Klopper 1999); emerging farmers do not use it extensively (Mukhala and Groenewald 1998; Vogel 2000; Archer 2003) and smallholder farmers use it even less (O'Brien *et al.* 2000; Ziervogel 2001; Patt forthcoming)[1]. One of the key reasons for smallholder farmers using it less than other groups is that they are least exposed to it (O'Brien *et al.* 2000). Addressing this inequity requires improving the dissemination process and increasing smallholder access to the forecast, which is the focus of another paper (Ziervogel and Downing forthcoming). One of the other key reasons for the forecast not being used by these groups is the limited capacity of smallholder farmers to respond to the forecast (O'Brien *et al.* 2000). This is not a well-researched topic.

Thus, the aim of this paper is to evaluate processes by which smallholder farmers filter seasonal forecast information and how they relate the forecast to their needs. This research should be fed back into dissemination plans and policy so that user groups can be more effectively targeted (Hammer 2000). If forecast development is dovetailed with users' needs there is more likelihood that response capacity will be improved and that it could aid pathways of sustainable livelihood development (Ortiz 1980; Weber and Sonka 1994; Hammer 2000). The paper first presents a framework for addressing the constraints associated with disseminating and integrating forecasts at the household level in rural southern Africa. This includes a discussion of the probabilistic nature of the forecast and how it could be integrated into household decisions among smallholder farmers. The current forecast product and communication channels within Lesotho are then described. The difficulty in understanding decision processes in response to information that is not pervasive requires an exploratory methodology to be used. The main section of the paper presents the methodology of participatory role play. The role-play exercise establishes how smallholder farmers could integrate the seasonal forecast into annual household decisions. Four key lessons learned from the role-play exercise are presented. In the final section, the discussion expands upon these lessons to provide more widely applicable lessons for the development of seasonal forecast applications.

## Decision processes

This paper starts from the premise that people are not always optimizing and rational and so idealized

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| Step 1 | Step 2 | Step 3 | Step 4 |
|---|---|---|---|
| **Dissemination effectiveness depends on** | **Integration of information based on filters** | **Decision options** | **Type of impact** |
| Method (radio, print media, personal communication)<br>Timing<br>Credibility | **External**<br>Suitable scale for response<br>Other households' responses<br>Availability of resources<br>**Internal**<br>More adverse to perceived loss than gain<br>Fit with endogenous knowledge<br>Perceived available options<br>Past experience | Alter crop management<br>Alter livestock management<br>Alter natural resource management<br>Alter household decisions<br><br><br>(data from role play) | Positive impact<br>Negative impact<br>No impact<br><br><br>Impact determines past experience filter the following year |

**Figure 1** Framework for assessing how forecast information is integrated into decisions

decision-making models will never be achieved (Clark and Marshall 2002). It assumes that smallholder farmers will not all make the same decisions in the same way and in isolation (Weber and Sonka 1994). It therefore builds on the notion of 'bounded rationality'. The decisions farmers make are constrained or bounded because they do not occur with perfect knowledge of all other variables, but rather are shaped by limited knowledge, cognitive capacity, resources and the structure of the environment in which decisions occur (Simon 1957). These factors are seen as filters in determining what the response to new information will be.

Figure 1 provides a framework, consisting of four steps, for addressing the problem of forecast integration into rural household decisions (Fischhoff 1994). The first step is the delivery or dissemination of the forecast. If the household receives the forecast, the second step occurs where filters are applied that either encourage or discourage further consideration of the forecast. The strength of the filters in Step 2 will depend on the perceived importance and potential use of the information and how the new information, if it is of value, is integrated into current knowledge and existing cultural frameworks (Agrawala and Broad 2002). The filters may be external or internal. External factors are those outside an individual's control. For example an external filter might be the importance placed on the forecast by other village or family members, or the limited resources that might constrain the number of available options. The internal filters are more a function of individual preference and perception. For example, the internal filter may depend on how the seasonal forecast conflicts or concurs with individual traditional climate forecasts and local environmental indicators. Cultural factors, for example, will determine how individuals view concepts of future and prevention. These concepts often have different manifestations in different cultures and so will determine the perceived need for adapting to the seasonal climate. Other internal filters might be a function of the perceived options that exist for

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responding to the forecast that are based on the history of decision making, the perceived riskiness of using the information and the success or failure of past decisions (Weber and Sonka 1994). If the filters allow for the forecast to be adopted, then in Step 3 decisions need to be made on how to use the information. Finally, in Step 4 the impact of the decisions made in response to the forecast are assessed. These can be positive, negative or no impact. The results from Step 4 (the impacts) become one of the filters for future years.

The numerous constraints associated with using the forecast can be broadly separated into problems of forecast delivery, which are those in Step 1, and problems with using the forecast, which are associated with Steps 2 and 3. The problems in Step 1, which are those of dissemination, include the mechanisms for delivery, the timing, and the message that is conveyed. The way in which the forecast is disseminated, such as through the radio, printed media or through people, influences how people react to the information and the quality of information that is passed on. The lack of efficiency in delivery often creates a problem. The time delay from when the forecast is issued to when the forecast reaches the users is known as the lag-time (Stern and Easterling 1999). The forecast might be issued in November but may only reach people a month later when they have already planted their crops and so the forecast is of little use to them. This lag-time is consequently too long. The credibility of the forecast, built on the accuracy of past forecasts and the forecast communicators' reputation, can also restrict uptake (Patt 2000; Patt and Gwata 2002). Improved credibility may depend on improved forecasts and better data or may depend on better communication that emphasizes the probabilistic nature of the forecast. These constraints are important, but addressing them relies on intervention before reaching the users. Once the information does reach the users, other constraints arise that depend on the user profile (Nicholls 2000) and these are reflected in Step 2.

The external filter of scale suitability arises out of climatologists and forecasters' imperfect knowledge of the physical environment. Imperfect knowledge and the chaotic nature of the atmosphere result in a probabilistic climate forecast rather than a spatially and temporally deterministic forecast (Murphy *et al*. 2001). This produces a coarse spatial resolution, which means that only a few different forecast zones cover one country, and a coarse temporal resolution, which means that the total rainfall is provided for one to three months but the expected distribution of rainfall within that time period is not given. This information might be appropriate for some users and not for others. The level of utility will depend on the intended use of the information (Blench and Marriage 1998; Orlove and Tosteson 1999). For example, a water manager might find the forecast to be detailed enough to be incorporated into a management plan of a catchment area for the season. A smallholder farmer, however, might decide that the forecast is not salient because it does not specify when the rains will start and how many days of rainfall there might be. Alternatively, that farmer might want additional information that will decrease the uncertainty associated with the season even if they are not going to change their operational decisions in response to the forecast. In this case they might be content with the existing spatial and temporal scale.

The internal filters, dependent on personal objectives and individual decision-making characteristics, will determine how the probabilistic nature of the forecast is accepted. According to Knight (1921) and followed up by Kahneman and Tversky (Kahneman and Tversky 1979 1986; Tversky and Kahneman 1991) among others, a calculated uncertainty equates to risk because the degree of uncertainty is specified. If the uncertainty was not specified, the risk would not be quantifiable and so would remain simply that, an uncertainty or a danger. Even though the forecast translates the uncertainty of the amount of rainfall expected in a season into a probability, so that it is no longer an undefined quantity, people still have difficulties dealing with probabilities. This has received little attention in the forecast literature (Nicholls 1999). It is important for disseminators of forecasts to understand this so that they can address the difficulties appropriately. If the forecast was designed as a 'quantification of the danger of climate variability' for example, it might allow users to accept the forecast more easily.

How do issues of dealing with probabilities relate to the use of forecasts by farmers? Part of the problem stems from the cognitive illusions or biases that people have (Nicholls 2000). These biases are often a product of heuristics or simplified strategies that people rely on when making decisions. For example, 'prospect theory', established from Kahneman and Tversky's work that was fuelled by their interest in whether people make risky or conservative decisions, showed that people do not view loss and gain equally (Kahneman and Tversky 1979). People rely on heuristics rather than statistics. For many people, the first priority is not to lose and the second priority is to gain. Some people are more conservative about actions that will lead to loss than to gain, whilst others are more

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risk-taking. Therefore it is important to establish the profiles of users, as stressing possible loss related to forecasts might motivate some types of users more than framing the same decision in terms of possible gain. Some farmers might be more risk-averse than others and it may be a function of wealth, education or age among other factors (Weber and Sonka 1994). Not only is the forecast probabilistic but also many of the other social and environmental aspects that constrain or enable decisions are uncertain. The external filter of resource availability is linked to the internal filter of perceptions of uncertainty. A farmer in a remote rural area may think of changing the type of crop planted in response to the forecast condition. The seed for that crop may not be available, or if it is, the farmer may not be able to access draught power when needed and so may have to wait an unknown length of time, or may have to wait for availability of labour that may vary depending on the health of the workers. All these other variables mean that using a forecast cannot be directly translated into probability of a good or bad yield. If, on the one hand, a below normal rainfall forecast is associated with a little rainfall and a poor harvest, people might be more hesitant to adopt strategies in response to information that they equate with loss rather than responding to an above normal forecast that they associate with a good harvest and gain. If, on the other hand, they also associate above normal conditions with a poor harvest they might be hesitant to use the forecast information to change their decisions at all.

The socio-economic and physical environments are underlying filters that affect how farmers will respond to new information about the climate as well as determining how the information is disseminated and the impacts that will occur. Southern Africa is an inherently risky environment to try and subsist in both because of the variable climate and the uncertain political and economic environment (Rowland 1993; Scoones and Thompson 1994). The climate variability is high primarily due to the rainfall variability, both within the season and between seasons (Rowland 1993). The effectiveness of the rainfall varies depending on the intensity, topography and ability to harness the water. Vulnerability to climate variability is therefore not uncommon as it is a risk most dryland farmers face regularly. Farmers are used to dealing with a risky environment. Their planting strategies have been developed to spread the risk of germination or crop growth failure (Hazell *et al.* 1986). Land is usually prepared before the rains start and planting occurs afterwards or at stages before and after in order to spread risk (Rowland 1993). Information that may decrease riskiness is therefore a potentially exciting prospect as avoidance of risk is a pervasive concern. Some farmers, however, might feel that probabilistic information is too risky to use because of the lack of certainty.

If enough importance is attributed to the forecast, decision options are considered (Step 3). Even if farmers accept the forecast they may decide not to change their short-term decisions, but as Adams and Mortimore state, contributing to increased longer term understanding is also important:

> Farmers . . . do not simply respond to exogenous and endogenous change reactively within each growing season. Their decisions, when considered in sequence, are cumulative and purposive and have, therefore, a longer term significance which reflects their understanding of longer term and larger scale changes.
>
> Adams and Mortimore 1999, 10

If they do decide to change their decisions, they may decide to adapt existing practices or adopt new practices. These decisions are primarily a function of risk management (Hazell *et al.* 1986), although there is little distinction between farm management and risk management (Fleisher 1990). As climate is one of the biggest risks for smallholder farmers (Gibberd *et al.* 1996) it is likely that many farm decisions could be affected by new information on what the rainfall might be. The decisions might be short-term tactical decisions, such as changing crop management or input supply choice, or long-term strategic decisions, such as crop rotations or natural resource management (Carberry *et al.* 2000). It is the details of these decisions that are uncovered in the role-play exercise.

Step 4 is the last step in the sequence and is not addressed in detail in this paper. Only once the filters and decisions are understood and the forecast applied can the impact be accurately assessed. The research tried to assess what farmers thought the impact of the forecast would be and how a false forecast would affect them, but it was not dealt with in enough depth to make a significant contribution to the potential impact of the forecast on rural households. An understanding of Steps 1–3 provides a good starting point for anticipating potential impacts (Ziervogel and Calder 2003). Further research on forecast use will uncover the realities of the impact of forecast adoption.

## The seasonal forecast and its place in Lesotho

Lesotho is landlocked by South Africa and covers an area of 30 000 $km^2$ (Ministry of Natural Resources 2000). The environment is particularly

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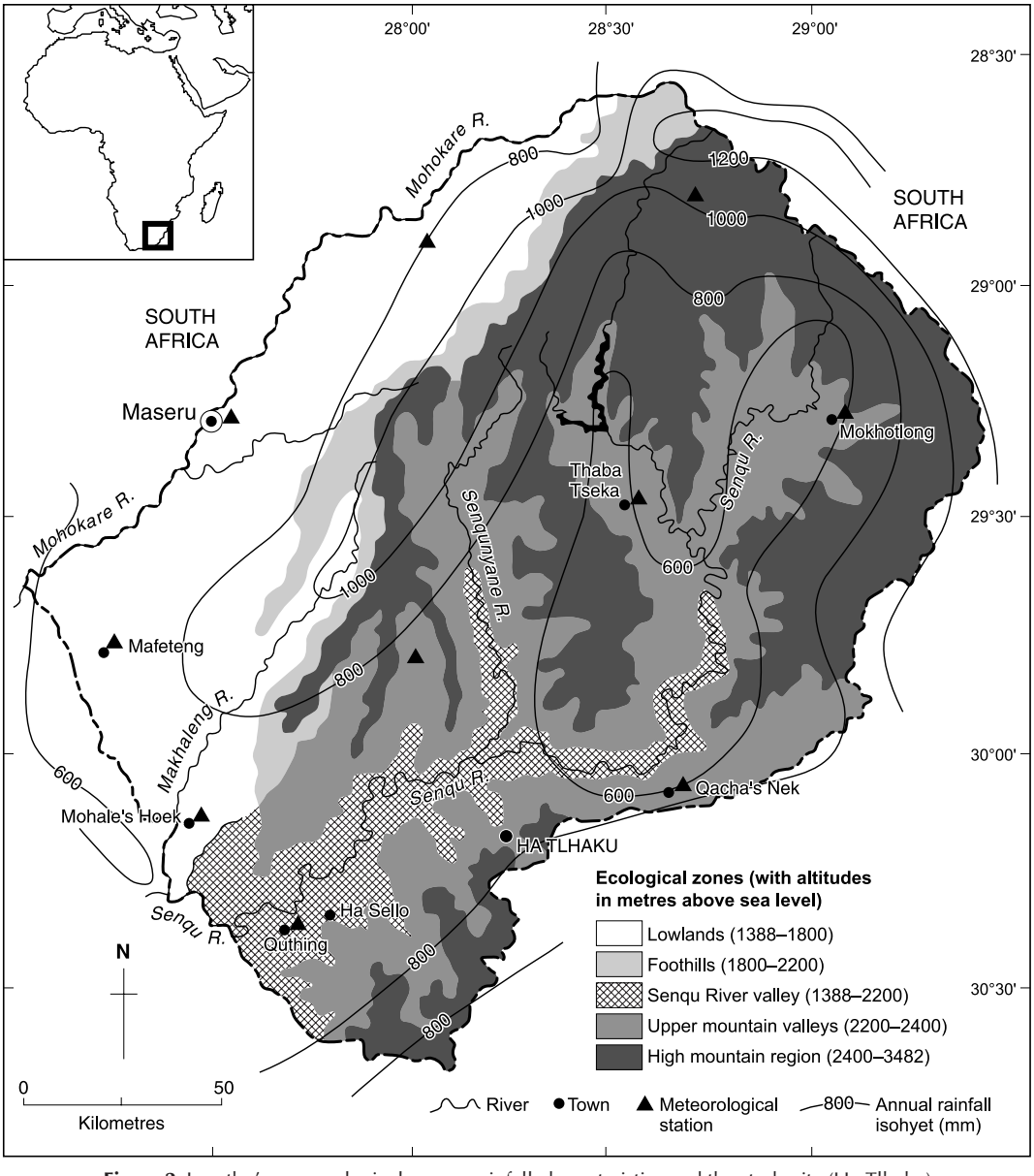



**Figure 2** Lesotho's agroecological zones, rainfall characteristics and the study site (Ha Tlhaku)

harsh because of its terrain and climate. The altitude ranges from 1388 to 3482 m.a.s.l. The mean annual rainfall ranges from 426 mm in the lowlands to 1097 mm in the highlands, with 85% falling in the summer months of October–March. Snow occurs annually in the mountains and frost is common across the country (Chakela 1999; Hyden and Sekoli 2000). The country is divided into four agro-ecological zones as seen in Figure 2: the lowlands in the western portion of the country rising to the foothills, and to the highlands in the east. The area adjacent to the Senqu river in the south of the country is often considered a separate zone. The highlands account for 60% of the land cover and have the lowest concentration of people.

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The Basotho[2] have been living in the mountains for about 200 years (Ambrose 1976; Chakela 1999), which is not a long habitation compared with many African groups. In these 200 years pastoralism and agrarian activities have supported a large proportion of households, as they did before moving to what is now Lesotho. Although there has been an increase in urbanization, agriculture still supports the majority of the population in Lesotho despite the myriad of challenges faced (Gay and Hall 2000; Turner *et al.* 2001). Agriculture is the obvious alternative to paid employment as everyone is entitled to land by law, labour is readily available and seed is usually kept from the previous year's harvest.

Lesotho was chosen as an appropriate country in which to undertake this research, as it is small enough to investigate the national operations, yet has many marginal groups that are highly dependent on rain-fed agriculture (Chakela 1999). The research is also complemented by other ongoing work in Lesotho that focuses on poverty and livelihood security (Calder *et al.* 2000; Gay and Hall 2000; Turner *et al.* 2001).

Lesotho Meteorological Services (LMS), housed in the Ministry of Natural Resources, are responsible for the development and dissemination of the seasonal forecast. Annually, since 1997, a regional climate forecast meeting, known as SARCOF (Southern African Regional Climate Outlook Forum), has been held in September in one of the southern African countries (Basher *et al.* 2001). At the meeting a consensus forecast is produced for the region. LMS then adapts the information to fit with their statistics and presents Lesotho's forecast at a National Workshop held in Maseru, the capital, at the beginning of October. Representation at the workshop is biased towards government department personnel, but there are usually a handful of private farmers, media representatives and others present. No radio programmes deal explicitly with the forecast although it is mentioned on the radio on a few occasions. The result is that not many people know about the seasonal rainfall forecast and those that do, only hear it after the workshop. The first rains start around October, so those who want to prepare for the season often do not have an adequate lead-time to factor the forecast into their decisions.

Seasonal forecasts are issued at the national level. Ideally they should pass through the various intermediaries to the end-users. This chain is seldom completed (Ziervogel and Downing forthcoming). LMS has a good relationship with the Ministry of Agriculture (MoA) and they expect them to pass the forecast on to their field staff. The MoA heads of departments, who are located in Maseru, receive the forecast from the October workshop and say that they send the information on to the District Agricultural Offices (DAOs), but none of the Extension Agents, who are under the DAOs, seem to know about the seasonal forecasts. Some of them think that they have heard what the rains will be like from the radio, but they are certainly not trained to disseminate the information to the village members they interact with. This is typical of the poor channels of information dissemination and limited support that seems to filter down to the farmers.

If the forecast is to reach end-users, the chain of dissemination needs to be strengthened. Before this is done, it is important to gauge whether the forecast will be accepted (in its present form and with the variable skill) and adopted when it reaches the end-users. There is more chance of it being adopted if individuals are able to assimilate the new knowledge with their existing knowledge (Eakin 1999; Stern and Easterling 1999). This is likely if the information resonates with previous experience and if the dominant cultural framework supports the integration of scientific information (Agrawala and Broad 2002).

## Role-play exercise: local perceptions of the seasonal forecast

An example of how marginal groups presently negotiate climate variability and how they might change their decisions in light of additional information is provided by the case study of Ha Tlhaku. The response from the role-play exercise provides a means of assessing local applications and implications of seasonal climate forecasts by drawing on the current perceptions of seasonal decisions, and of how a seasonal climate forecast could be integrated into existing knowledge systems.

Ha Tlhaku (see Figure 2), a village composed of about 90 households (with each household having between two and fifteen people), is located in the south of the country in the mountains of the Qhoali valley. It is a two-hour drive from the district headquarters, Quthing, and half an hour from Mphaki, the mountain service centre, where there are input suppliers and a MoA Farmers' Training Centre. It is a successful agricultural village in terms of crop varieties and yields, which are better than in nearby villages. This can partly be attributed to favourable natural resource endowment and partly to a well-respected enthusiastic chief (who also governs nearby villages) who ensures that his subjects get as much help as possible.

Ha Tlhaku is a traditional village with a high dependence on agriculture and livestock (Mohasi and Turner 1999). There are three projects that operate in the village: CARE's TEAM (Training

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of Environmental and Agricultural Management) project which facilitates experiential learning sessions on agricultural techniques that use locally available inputs (Calder *et al.* 2000) and through which this research was introduced to the village; the Durham Link project which focuses on tree-planting and sale; and the Mphaki project which promotes livestock production and development. Previous fieldwork showed that there are a number of environmental indicators that people have used in the past to understand local weather, such as the angle of the moon and the call of a certain bird, but few of the indicators refer to seasonal climate patterns (Ziervogel 2001).

### *Methodology*

A role-play game was developed in order to understand how participants view and might use seasonal forecasts. Previous fieldwork used surveys to establish how farmers could use the seasonal forecast, but the results were not detailed enough to give an understanding of how the forecast could be integrated into decision making. The role play, therefore, was developed out of the surveys in an attempt to contextualize the forecast information and put people in a more 'normal' decision-making context where they could think about their households and the range of decisions they make each year at the beginning of the rainy season. Decisions were characterized on an annual basis for the start of the season and were sequential because decisions made in one year usually depend on what has happened in the previous year and on the future vision or strategy of the household (Weber and Sonka 1994).

The role-play game was played six times within one week in Ha Tlhaku. Two of the games were played with individuals, one with a male group, two with female groups and one with a mixed group. The groups consisted of between three and six people with 22 people participating in total. The participants had to own or work in fields in order to participate. Typically, the chief would send out a messenger in the morning and ask women to participate, for example. If people did not arrive, then a household would be visited and asked if they would participate. Two Sesotho-speaking agricultural graduates acted as Extension Agents and facilitated the game; one asked the questions in Sesotho and the other one recorded the answers in English. The exercise, which usually lasted for about two hours, was done in the month of September 2001, which is the month preceding the start of the rainy season.

Each game consisted of three rounds played over three imaginary years. The first round took place in 'the present year', which was 2001. Participants were asked to draw their village on a large sheet of paper. Everyone was encouraged to make at least one mark on the sheet. They were then asked to look at what they had drawn and say what decisions they would have to make in the season with regards to their resources, including fields, livestock, water resources and general household decisions. In the second round they had to pretend that it was the same time the following year, 2002. A forecast for below normal rainfall was read out in Sesotho as though it were a radio broadcast. They then had to reassess their decisions in the light of the new information and state what decisions and activities they would undertake for that year. The facilitators asked a series of questions concerning how they would explain the forecast to their neighbour and what advice they would give them on how to prepare for the season. They were also asked if, and how, they would act on the radio forecast or whether they would wait and see what the rains were like. A third round was played for 2003, but this time the forecast was for above normal rainfall[3] and the same questions were asked as in the second round. After the three years of decision-making responses a final round of questions was asked about preferred methods of forecast receipt, the impact of a false forecast, whether poor and better-off farmers could benefit equally, and whether they would use the forecast if they heard it on the radio next year.

### *Results*

The role-play exercise provides insight into what could be done in response to forecasts if information reached the villagers, although it builds on speculative rather than actual behaviour. The various rounds highlight whether farmers thought that they would like to receive a forecast and, if so, what responses and decisions could be made if farmers accept the forecast and had the resources to act as they wished.

Identifying the type of decisions that forecasts could be used for was one of the objectives of the exercise. Table 1 shows a selection of the suggestions participants provided on how they would advise their neighbours to act on the given forecast. These were sometimes similar to what they said they themselves would do in response to the forecast and sometimes different, as even if they did not think they would change their decisions they still had suggestions as to how their neighbours might use it.

The first finding is that there is no shortage of ideas on how seasonal forecast information might be put to use and, more importantly, many of the

**Table 1** Farmers' suggestions on how to adapt to below and above normal rainfall

| Advice given to other farmers about what might be done if | |
|---|---|
| Predictions for below normal rainfall | Predictions for above normal rainfall |
| Plant drought resistant crops<br>Sow less maize and wheat | Grow more vegetables as cash crops<br>Plant earlier |
| Reduce density of field crops | Increase crop densities |
| Irrigate if the streams still have water: dig furrows next to fields to stop wilting | Increase sharecropping[1] as it results in greater yields (more labour and more land) |
| Collect water and pour over plants | Be aware of diseases and pests for crops and animals |
| Plough lucerne and teff as animal feed since they survive harsh conditions | Plant different crops to neighbours so that there is a market |
| Reduce the number of animals which cannot survive the harsh conditions (by selling) | Practise winter breeding since there will be water and feed for the sheep and goats |
| Take animals to the *mofebong* (rangelands) where there is better grazing and water | Build animal shelters |

[1]Sharecropping: one party supplies the land and the other the inputs.

ideas do not require a lot of resources and so are appropriate for a range of livelihoods. Many of the suggestions given in Table 1 relate to the management of field crops. Although livestock are a high priority for the Basotho, there do not seem to be as many things that can be done to maximize utility or minimize risk for the livestock in relation to the climate. While one of the suggestions for responding to below normal rainfall was to sell animals rather than have them perish in the harsh conditions, this would only occur under extreme circumstances, as livestock are not easily sold because people see them as a status symbol (Ferguson 1985). There were not many suggestions given when participants were asked how the forecast information might be used to manage household decisions. This is not surprising, as it is harder to relate the impact of climate to the indirect effects on the household, as opposed to the direct impact on resources. When people were probed on whether they would send their children to school in drought years, many agreed that they might have to pull the children out of school for that year, but it was not an immediate response to the forecast.

The second finding is the preference for above normal forecasts. Most of the participants said that they would respond to a forecast for above normal rain but would ignore the forecast if it predicted below normal rain. This did not stop them giving suggestions as to how they might use the below normal forecast if they did believe it. One farmer said, 'A useful forecast is one that predicts better rains as that will benefit us as we can grow more vegetables'. A more sceptical farmer said, 'The forecast will interest us if it is good but we will still assume normal conditions of previous years will prevail'. They said that if the ground were so dry that it was too hard to plough, it would make them change their activities as they would have to wait for rains before preparing their fields. This seems to indicate that the villagers are more likely to use the forecast to maximize good conditions but are not necessarily prepared to believe the predictions of what they perceive to be bad climatic conditions. The reason for this could be that they do not want to acknowledge the fact that conditions will be bad, especially when many households do not have any form of insurance, whether financial or resource capital, and there seems to be little they do to prepare for, or curtail, climate-induced negative impacts. A poor farmer commented that 'We cannot sit back and rely on a radio message as this is our only way of survival'. This is important if one acknowledges that Early Warning Systems and El Niño warnings alert people to the extremes rather than to the average conditions. Possibly it is the ability to maximize good conditions that would help marginal groups improve their livelihoods rather than information that would help them survive the hazards.

The third finding is based on a trend that was exhibited in the answering of questions 'over the years'. At the end of each round people were asked whether they would use that year's forecast

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information or not. 'Over the years' participants seemed to be more open to using the forecast information. In response to the first predictions of below normal rainfall everyone from all the groups, except one better-off farmer, said that they would not use the forecast. In response to the second prediction of above normal rainfall, about half of the people said they thought that they would act on the forecast information that year. By the last round, when they were asked if in reality they would use the forecast information if they heard it on the radio the following year, they all said that they thought they would. This could be due to the bias towards above normal forecasts but that would not explain the increase in numbers when asked whether they thought they would use it in reality if they heard it the following year. One explanation is that learning occurs over the years as people become more familiar with the type of information provided. The third time they were asked whether they would use the forecast, they had given examples as to how they might use it and became aware that they had many ideas on how to operationalize the information. Using a forecast no longer seemed foreign at that stage, unlike the first time when they might have thought that they did not know how to use the information.

The fourth finding relates to how villagers would like to receive the forecast. The most popular method of forecast receipt was from the Extension Agents, and specifically the CARE extension staff who visit the village regularly. Other preferences were to hear the information from the chief, through *pitsos* (community gatherings) and from Sesotho newspapers, although someone commented that these are often out of date by the time they reach the village, if they reach the village at all.

Radio was often mentioned as a way of receiving the forecast although it had constraints. Many poorer farmers said that radios were not always available and that even if they owned radios they could not always afford batteries so would not hear up-to-date information. An important constraint that many mentioned was that they cannot ask questions when they do not understand something on the radio. They said that feedback was very important when they were faced with new information. This is reflected in the role-play exercise, as acceptance of the forecast grew when they became more familiar with it. The participants were able to discuss options among themselves and with the Extension Agents. In the game the Extension Agents were not allowed to offer initial suggestions but did comment on participants' suggestions if asked, since, in reality, it was the individuals' decisions that were important.

In the final round participants were asked whether they thought farmers with more resources could utilize the forecast more successfully than those with few resources. The objective was to establish whether the villagers had the same perception as many researchers who suggest that poorer people are restricted in responding to the forecast because of their limited resources and lack of ability to control their environment (Hulme *et al.* 1992; Vogel 2000). In this exercise the majority of farmers said that they thought that both would benefit equally as they farm in the same conditions. Some people thought that poorer farmers need any help they can get, whereas others thought that the better-off farmers would benefit more as they have better access to information as well as better farming methods. A realistic expectation of who might use the information most comes from an Extension Agent's comment:

> The people who are able to establish a link between the forecast and how they do things and how they need to change are more likely to use it and those will be the better-off farmers; they are also the ones who want to make more money and are more willing to take risks than the poor.

## Discussion

The role-play exercise offers four key lessons concerning the integration of the forecast into the livelihoods of smallholder farmers in Lesotho, although it is based on hypothetical rather than actual response to the forecast. Although these lessons emerge from the southern African environment, they have implications for forecast integration into household decision making for other developing regions. For the lessons to be useful in other regions, it is imperative that the local environment be used to contextualize potential forecast development as existing structures and institutions will determine local response.

### *Forecast response options exist at the household scale for all levels of resource endowment*

Smallholder farmers do not require many resources at the household level in order to respond to the forecast. Options such as changing cropping densities can be done by all households and many can change crop type by bartering with their neighbours, as is common practice. Changing to drought resistant varieties might only be an option for households who can afford to buy new seed, unlike most of the households that keep their seed from the harvest to plant the following year. Everyone can become more aware of the possibility of

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increased disease incidence, but response options may be limited to those with the means to do so. Acknowledging this range of response options is an important starting point for further work. The implications are that resource-poor farmers should not be excluded from dissemination networks for the reason that they do not have the resources with which to respond. It does not mean that resource-poor farmers will use the information even if they do receive and understand it, as there may be other reasons for the information not being used, as outlined in the 'Decision processes' section. The socio-economic or cultural environment may constrain the adoption of the forecast, as there may be a lack of 'institutional fit' where the information given and the users' needs do not meet up (Orlove and Tosteson 1999). There might also be more specific reasons such as not wanting to use probabilistic information or not trusting the information, or the information may contradict endogenous climate information (Nicholls 1999; Orlove and Tosteson 1999; Patt and Gwata 2002).

The types of decisions that people mentioned they might undertake in response to the forecast were mostly short-term tactical decisions such as 'sowing less maize and wheat', 'planting earlier' or 'protecting the livestock' rather than long-term strategic decisions. These decisions are a form of agricultural risk management. The focus on short-term decisions is expected when the information is new since, over time, when these decisions seem an appropriate way of responding to the forecast, more long-term decisions can be made in response to the forecast depending on the risk preference of the decisionmaker (Carberry *et al.* 2000). Some people mentioned that they might keep their children from school in drought years, as they could put the money for school fees towards feeding the family. Although these suggestions were made in response to the hypothetical forecast, it seems unlikely that farmers would make a serious decision like this if the forecast reached them next year and they had not 'tested' it for themselves before. The risk is potentially too high. If they did and the actual rainfall was not as they expected from listening to the forecast, the implications could be serious. These types of decisions should not be made without people understanding the complexities of a probabilistic forecast as it could make them more vulnerable. This was seen in Zimbabwe in 1997 when farmers were denied bank loans because the predicted drought did not occur (Orlove and Tosteson 1999; Patt 2000; Agrawala and Broad 2002). The forecast provides an opportunity for better managing risk, although the impact of who will benefit is not always clear.

*Selective adoption of forecast information occurs*

In Ha Tlhaku, the forecast for above normal rainfall was preferred to the forecast for below normal rainfall and people said that they would listen to one and not the other. This can be attributed to the user profile. Different users will have varying levels of desire and ability to take risks and it is important to consider these user characteristics when addressing dissemination strategies. For example, a forecast for below normal rainfall could be more suitably targeted if the disseminators realize that it will not be considered in an equal light to the above normal forecast. This fits with the prospective theory that says that equal risk of loss is more cautiously followed than equal risk of gain. Farmers were hesitant to use the forecast that they thought would lead to a poor harvest and were more willing to use the forecast they thought would lead to gain. This is not obvious, as disseminators might have expected people to try and decrease negative impacts and so use the forecast to avoid loss. If farmers did this, it would require an acknowledgement of a potentially poor season, which these farmers did not want to do. Other farmers, that are more risk seeking, might respond differently.

Forecast response options are presented in Table 1. These responses have been reassessed in Table 2 to classify what the impact of a false forecast would be if the responses had been undertaken. The classification is loose but is determined by asking if the impact would be big or small if the forecasted climate was different to the actual climate. In the case of there being a small impact, this means that farmers might have invested inputs, such as setting up irrigation, but this would not necessarily have a dramatic negative effect. The impact of a false forecast was worse when farmers responded to the forecast for below normal rainfall than when they responded to an above normal forecast. This provides another possible reason for why farmers might have preferred the above normal forecast over the below normal forecast.

These findings are important for work on Early Warning Systems, where extremes such as drought are flagged. This research indicates that farmers may be hesitant to respond to extreme warnings. This is where the seasonal forecast could have an advantage over previous warning systems. The forecast is available every year and not just in extreme circumstances. This means that systems will be in place if an emergency situation arises. It also means that people can use the forecast to make the most of normal or good years rather than just decrease the impact of hazards. This allows for an understanding of the response options to build

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**Table 2** Impact of false forecasts on livelihoods if the forecast is followed

| Response to below normal rainfall | Impact of false forecast | Response to above normal rainfall | Impact of false forecast |
|---|---|---|---|
| Plant drought resistant crops | Big | Grow more vegetables | Big |
| Sow less maize and wheat | Big | Plant earlier | Big |
| Reduce density of field crops | Big | Increase sharecropping | Small |
| Set up irrigation | Small | Increase crop densities | Big |
| Collect water and pour over plants | Small | Plant different crops to neighbours | Small |
| Plant animal feed | Big | Be aware of diseases | Small |
| Reduce the number of animals by selling | Big | Practise winter breeding | Big |
| Take animals to rangelands | Small | Build animal shelters | Small |

up over the years and so encourages learning, which is critical when aiming to facilitate adaptation.

*Learning is experienced through repeated forecast dissemination*

In the first round, farmers sometimes said that they had traditional forecasts that they would trust more than the seasonal forecast. By the last round this was less common as they had learned that the seasonal forecast could be used as additional information and did not have to compete with traditional forecasts. This lesson is important for scientists and disseminators, who need to be reminded that they cannot expect forecast adoption at the initial inception phase, but rather that it is constant exposure that will increase the usage (Klopper 1999). This is more likely to happen if feedback mechanisms, which integrate the users' needs into future forecast products, exist. In Ha Tlhaku the traditional forecast did not seem to conflict with the seasonal forecast. Nevertheless, it is important to appreciate the role that traditional forecasts play as they can be used as an analogy to help forecast uptake (Eakin 1999). It is this sensitivity to the cultural environment that is imperative for appropriate targeting and will advance the uptake capacity.

Seasonal forecasts have been operational in Lesotho for five years, but there does not seem to be much use or awareness of the forecast at the national level and even less at the local level. If learning is to occur, the forecast needs to have a higher profile so that people are aware of what it is and how they can use it. Improved awareness of the forecast will not happen if responsibility stays solely in the hands of meteorological institutions. Cross-sectoral communication and development, which links the impact of the climate and the forecast across the various sectors, needs to be strengthened and used to identify local constraints (Gibberd *et al.* 1996; Broad 2000). If there is support from both public and private institutions there will be a richer context of information delivery and accelerated learning (Roncoli *et al.* 2000).

*Interactive dissemination methods are preferred*

The radio is often proposed as a method that could be better utilized for disseminating forecasts, as it can reach a lot of people in a short space of time (Stack 1998; Chapman *et al.* 2003). Although farmers are not averse to radio broadcasts, they did say that if they were learning about a new product or technique they preferred to see demonstrations of what could be done. Not only does this give them the chance to decide whether they want to use the information or not, it provides an opportunity for asking questions (Hagmann *et al.* 1999). The problem with the radio is that it does not have the ability to integrate users' needs and perceptions easily (Omosa 1998). This may result in low forecast adoption rates and a lack of understanding and therefore utility. As stressed in the earlier section, it is important for users to understand the probabilistic nature of the forecast as this can determine how much trust is placed in the forecast. If the probabilistic nature is not accurately conveyed and the forecast is different to the outcome, users could blame the disseminators of the forecast rather than appreciating at the time of use that they are taking a risk. For this reason reliable methods of information delivery should be a priority.

Receipt of the forecast from Extension Agents was one of the most popular options for receiving the forecast. In response to this, staff from CARE Lesotho were interviewed to establish what role they thought they could play in disseminating the

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forecast. The CARE staff said that at present the information they pass on includes training village members to run modules that deal with a range of issues from managing the soil using mulching, improving fertility by using organic matter, ash and compost to adjustment of field design (such as raising or lowering plots) and marketing advice. They emphasized the fact that they did not tell farmers what to do but rather ran experiments so that farmers could see what happened and could subsequently choose what they thought would work for them. The CARE staff felt that they could play a role in disseminating the forecast. Their suggestions included responding to below normal rainfall forecasts by using water conservation methods or undertaking activities that decrease water logging in years with above normal rainfall forecasts. Many of them were concerned that farmers would not understand the fact that forecasts were probabilistic, although they felt that the forecast should still be passed on despite this. The emphasis should be on ensuring that farmers are aware of options for responding to the forecast and, more importantly, that they take responsibility for using the forecast upon themselves and so are active in the decision-making process. When asked if they thought farmers would use the forecast information, most of the CARE staff said some would and some would not. One comment gives an insightful view of what is likely to happen:

> Traditionally we [Basotho] are not eager to take risks. Last year there was good rain but early frost so this year even though there was good early rain and people could have planted earlier for a better harvest they have still waited until November to make sure the rains have continued. It is good to take risks but you need to be careful of the scale so it would be unwise to take big risks on all fields.

Although the CARE staff see the potential for using the forecast and a role for themselves in disseminating the forecast, it is the responsibility of the LMS to seek these alternative channels to improve forecast dissemination.

### Lessons for future applications research

The role-play exercise also provides lessons about methodology. Although participatory research is not new, there do not seem to be many examples of role-play exercises. Assessing the strengths and weaknesses of this approach enables it to be expanded on and developed for future research. The strength of the exercise lay in the ability to go beyond taking a snap-shot of how people think they would react by encouraging a more nuanced understanding of the problem. This was achieved by providing more background to the problem and the local context before participants responded to questions. It also allowed for group interaction, which is something that happens when decisions are made in the village but may not be encapsulated in an individual interview.

There are a number of weaknesses associated with the design of the role-play exercise. The first is the limited ability to capture the impact of responding to the forecast. Although participants were encouraged to think about the previous year's forecast and how the outcome would affect them when making decisions in the following year, no mechanisms were in place to enforce them to realize and record the impact. The final round anticipated this and asked the question of how a false forecast would impact on households. The farmers responded by saying that if the forecast were different to the outcome they would have poor yields and would not trust the forecast, in particular the radio forecast, in the future. They said that inputs would have been wasted. Only a handful of participants linked the loss of crops to having 'nothing to eat'. Taking it beyond these obvious responses to whether a false forecast would make farmers more vulnerable or not, and whether a correct forecast would improve pathways of sustainable livelihoods, requires further probing of the issue, and perhaps the impact will only be seen when the forecasts are disseminated and reach these groups. Another aspect that requires caution is related to both the strength and weakness of participatory research. As was seen with the 'learning over the years', people's perceptions of forecasts changed because of the role-play exercise. This could lead to further enquiries and better awareness of the forecast or could result in disappointment if the research is not followed up. Another comment on participatory research comes from the observation that it was the more educated farmer groups who were prepared to draw the initial diagrams of their village and resources and the illiterate ones who refrained. Participatory research is advocated because of its ability to include all groups including the illiterate (Chambers 1994; Hagmann *et al.* 1999), but it was the illiterate who felt threatened by having to draw in this exercise (Nelson and Wright 1995).

Although this research has been undertaken in southern Africa it has implications for seasonal forecast applications in other developing regions. The lessons can also be applied to other types of scientific developments that could benefit marginal groups. The main points to consider are that dissemination has to be sensitive to user profiles, which include social norms, cultural preferences

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and institutional barriers. At the same time, existing resources and channels should be used where appropriate, but their limitations need to be identified early on, so that effective and efficient channels can be developed if necessary (Ziervogel and Downing forthcoming).

There are a number of ways for taking this research forward. One approach would be social simulation. A pilot agent-based model has been developed using the results of the role-play exercise to formalize how the forecast impacts on household decisions and subsequent production levels and village success. The research can also be developed to address how marginal groups might adapt to climate change (Adams and Mortimore 1999; O'Brien *et al.* 2000). Seasonal forecasts provide a unique analogy of addressing how climate information could be integrated into decision making on the medium-term scale of up to a year. This provides a stepping-stone for anticipating how longer-term climate information, particularly about climatic change, could be integrated into adaptation strategies. There are few other cases that provide this intermediary timescale which can be used to bridge short-term adaptation in terms of weeks and months, with longer-term adaptation on the decadal scale. For future research to enable targeting of forecasts to be more aligned with user needs and encourage pathways of sustainable development, it is necessary for actual decisionmakers to be involved in the research process.

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## Notes

1 Commercial farmers are those with large-scale production; emerging farmers are those farmers who were previously disadvantaged and are now starting to farm commercially, although their output is usually less than the traditional commercial farmers; and the term smallholder is used to refer to farmers who are trying to subsist from their land. Some smallholders are able to produce enough to live off and sell surplus, but some do not produce enough to feed the household for the year.

2 The people who live in Lesotho are known as Basotho. The singular is Mosotho and they speak Sesotho.

3 Most of the country is mountainous, so there is high runoff and flooding seldom occurs. A prediction of above normal rainfall does not therefore bear the same potentially threatening message that it might in many other parts of southern Africa where heavy rains are often accompanied by flooding.

## References

**Adams W M and Mortimore M** 1999 'Whose drought? Rain, crops and farm labour in the Nigerian Sahel' Paper presented at African Environments: Past and Present, Oxford

**Agrawala S and Broad K** 2002 Technology transfer perspectives on climate forecast applications in **Laet M D** ed *Research in science and technology studies: knowledge and technology transfer* volume 13 45–69

**Agrawala S, Broad K and Guston D H** 2001 Integrating climate forecasts and societal decision making: challenges to an emergent boundary organisation *Science, Technology and Human Values* 26 454–77

**Ambrose D** 1976 *The guide to Lesotho* Winchester Press, Johannesburg

**Archer E R M** 2003 Identifying underserved end-user groups in the provision of climate information *Bulletin of the American Meteorological Society* 84 1525–32

**Baster R, Clark C, Dilley M and Harrison M** eds 2001 *Coping with the climate: a way forward* IRI-CW/01/2 International Research Institute for Climate Prediction, Palisades, New York

**Blench R** 1999 Seasonal climatic forecasting: who can use it and how should it be disseminated? *Natural Resource Perspectives* 47 ODI, London

**Blench R and Marriage Z** 1998 The social and technical construction of weather: El Nino and other climatic events in sub-Saharan Africa *Natural Resource Perspectives* 31 ODI, London

**Broad K** 2000 'Practical and conceptual challenges to climate forecast applications' Paper presented at the International Forum on Climate Prediction, Agriculture and Development International Research Institute for Climate Prediction, New York

**Calder R, Mokhameleli S, Mahase M, Mohasi M, Lerotholi M and Khobotle M** 2000 *TEAM project monitoring and evaluation guide* CARE Lesotho, Maseru

**Cane M A, Zebiak S E and Dolan S C** 1986 Experimental forecast of El Niño *Nature* 321 827–32

**Carberry P, Hammer G L, Meinke H and Bange M** 2000 The potential value of seasonal climate forecasting in managing cropping systems in **Hammer G, Nicholls N and Mitchell C** eds *Applications of seasonal climate forecasting in agricultural and natural ecosystems – the Australian experience* Kluwer Academic Publishers, Dordrecht 167–81

**Chakela Q K** ed 1999 *State of environment in Lesotho 1997* National Environment Secretariat, Maseru

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**Chambers R** 1994 Participatory Rural Appraisal (PRA): analysis of experience *World Development* 22 1253–68

**Chapman R, Blench R, Kranjac-Berisavljevic' G and Zakariah A B T** 2003 Rural radio in agricultural extension: the example of vernacular radio programmes on soil and water conservation in N. Ghana *AgREN* 127 ODI, London

**Clark G L and Marshall J** 2002 'Decision-making: models of the real-world and expertise' Paper presented at National Association of Pension Funds, Edinburgh

**Eakin H** 1999 Seasonal climate forecasting and the relevance of local knowledge *Physical Geography* 20 447–60

**Ferguson J** 1985 The bovine mystique: power, property and livestock in rural Lesotho *Man* 20 647–74

**Fischhoff B** 1994 What forecasts (seem to) mean *International Journal of Forecasting* 10 387–403

**Fleisher B** 1990 *Agricultural risk management* Lynne Rienner Publications, Boulder

**Gay J and Hall D** 2000 *Poverty and livelihoods in Lesotho, 2000: more than a mapping exercise* Sechaba Consultants, Maseru

**Gibberd V, Rook J, Sear C B and Williams J B** 1996 *Drought risk management in southern Africa: the potential of long lead climate forecasts for improved drought management* Natural Resources Institute, London

**Hagmann J, Chuma E, Murwira K and Connolly M** 1999 Putting process into practice: operationalising participatory extension *Agriculture Research and Extension Network* ODI, London

**Hammer G L** 2000 A general systems approach to applying seasonal climate forecasts in **Hammer G, Nicholls N and Mitchell C** eds *Applications of seasonal climate forecasting in agricultural and natural ecosystems – the Australian experience* Kluwer Academic Publishers, Dordrecht 51–65

**Hazell P, Pomareda C and Avaldes A** eds 1986 *Crop insurance for agricultural development* John Hopkins University Press, Baltimore

**Hulme M, Biot Y, Borton J, Buchanan-Smith M, Davies S, Folland C, Nicholds N, Seddon D and Ward N** 1992 Seasonal rainfall forecasting for Africa. Part II: Application and impact assessment *International Journal of Environmental Studies* 40 103–21

**Hyden L and Sekoli T** 2000 Possibilities to forecast early summer rainfall in the Lesotho Lowlands from the El-Nino/Southern Oscillation *Water SA* 26 83–90

**Jones J W, Hansen J W, Royce F S and Messina C D** 2000 Potential benefits of climate forecasting to agriculture *Agriculture, Ecosystems and Environment* 82 169–84

**Jury M R** 2002 Economic impacts of climate variability in South Africa and development of resource prediction models *Journal of Applied Meteorology* 41 46–55

**Kahneman D and Tversky A** 1979 Prospect theory: an analysis of decision under risk *Econometrica* 47 263–91

**Kahneman D and Tversky A** 1986 Rational choice and the framing of decisions *Journal of Business* 59 S251–8, 285–300

**Klopper E** 1999 The use of seasonal forecasts in South Africa during the 1997/98 rainfall season *Water SA* 25 311–17

**Knight F H** 1921 *Risk, uncertainty and profit* Harper Torchbooks, New York

**Martin R V, Washington R and Downing T E** 2000 Seasonal maize forecasting for South Africa and Zimbabwe derived from an agroclimatological model *Journal of Applied Meteorology* 39 1473–9

**Mason S J, Joubert AM, Cosijn C and Crimp S J** 1996 Review of seasonal forecasting techniques and their applicability to southern Africa *Water SA* 22 203–9

**Maxwell S** 2001 Agricultural issues in food security in **Devereux S and Maxwell S** eds *Food security in sub-Saharan Africa* Institute of Development Studies, London

**Ministry of Natural Resources** 2000 *National report on climate change: Lesotho* Lesotho Meteorological Services, Maseru

**Mohasi M and Turner S** 1999 Land and livelihoods in southern Lesotho Programme for land and agrarian studies (PLAAS) and CARE, Lesotho

**Mukhala E and Groenewald D C** 1998 Experiences and perceptions of black small-scale irrigation farmers in the Free State *South African Journal of Agricultural Extension* 27 1–18

**Murphy S, Washington R, Downing T E, Martin R V, Ziervogel G, Preston P, Todd M, Butterfield R and Briden J** 2001 Seasonal forecasting for climate hazards: prospects and responses *Natural Hazards* 23 171–96

**Nelson N and Wright S** 1995 *Power and participatory development* Intermediate Technology Publications, London

**Nicholls N** 1999 Cognitive illusions, heuristics, and climate prediction *Bulletin of the American Meteorological Society* 80 1385–97

**Nicholls N** 2000 Opportunities to improve the use of seasonal climate forecasts in **Hammer G, Nicholls N and Mitchell C** eds *Applications of seasonal climate forecasting in agricultural and natural ecosystems – the Australian experience* Kluwer Academic Publishers, Dordrecht 309–27

**O'Brien K, Sygna L, Næss L O, Kingamkono R and Hochobeb B** 2000 Is information enough? User responses to seasonal climate forecasts in Southern Africa CICERO report no. 2000–03

**Omosa E** 1998 The use and application of various communication channels at local and international levels *Proceedings of a Workshop on Women in Agriculture and Modern Communication Technology* 30 March–3 April 1998 Tune Landboskole, Denmark (http://www.husdyr.kvl.dk/htm/php/tune98/12-EileenOmosa.htm) Accessed June 2002

**Orlove B and Tosteson J** 1999 *The application of seasonal to interannual climate forecasts based on ENSO events: lessons from Australia, Brazil, Ethiopia, Peru, and Zimbabwe* Institute of International studies, University of California, Berkley

**Ortiz S** 1980 Forecasts, decisions, and the farmer's response to uncertain environments in **Barlett P** ed *Agricultural decision making: anthropological contributions to rural development* Academic Press, New York 177–201

**Palmer T N and Anderson D** 1994 The prospects for seasonal forecasting – a review paper *Quarterly Journal of the Meteorological Society* 120 755–93

**Patt A** 2000 *Communicating probabilistic forecasts to decision makers: a case study of Zimbabwe* Belfer Center for Science

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and International Affairs (BCSIA) Environment and Natural Resources Program, Kennedy School of Government, Harvard University, Cambridge

**Patt A** forthcoming Trust, respect, patience, and sea surface temperatures: useful climate forecasting in Zimbabwe in **Alcock F, Cash D, Mitchell R and Clark W** eds *Information as influence: how institutions mediate the impact of scientific assessments on international environmental affairs*

**Patt A and Gwata C** 2002 Effective seasonal climate forecast applications: examining constraints for subsistence farmers in Zimbabwe *Global Environmental Change: Human and Policy Dimensions* 12 185–95

**Roncoli C, Kirshen P, Ingram K and Flitcroft I** 2000 'Opportunities and constraints to using seasonal precipitation forecasting to improve agricultural production systems and livelihood security in the Sahel-Sudan region: a case study of Burkina Faso, CFAR – Phase 1' Paper presented at IRI Forum on Climate Predictions, Agriculture and Development, New York

**Rowland J R J** ed 1993 *Dryland farming in Africa* Macmillan Education, London

**Scoones I** 1997 *Hazards and opportunities: farming livelihoods in dryland Africa, lessons from Zimbabwe* Zed Books, London

**Scoones I and Thompson J** 1994 Knowledge, power and agriculture – towards a theoretical understanding in **Scoones I and Thompson J** eds *Beyond farmer first: rural people's knowledge, agricultural research and extension practice* Intermediate Technology Publications, London 16–32

**Simon H** 1957 *Models of man* Wiley, New York

**Sivakumar M V K, Gommes R and Baier W** 2000 Agrometeorology and sustainable agriculture *Agricultural and Forest Meteorology* 103 11–26

**Stack J** 1998 Drought forecasts and warning in Zimbabwe: actors, linkages and information flows Working paper Department of Agricultural Economics and Extension University of Zimbabwe, Harare

**Stern P and Easterling W** eds 1999 *Making climate forecasts matter* National Academy Press, Washington DC

**Turner S, Calder R, Gay J, Hall D, Iredale J, Mbizule C and Mohatla M** 2001 *Livelihoods in Lesotho* CARE Lesotho, Maseru

**Tversky A and Kahneman D** 1991 Loss aversion in riskless choice: a reference-dependent model *Quarterly Journal of Economics* 106 1039–61

**Vogel C** 2000 Usable science: an assessment of long-term seasonal forecasts among farmers in rural areas of South Africa *South African Geographical Journal* 82 107–16

**Washington R and Downing T E** 1999 Seasonal forecasting of African rainfall: prediction, responses and household food security *The Geographical Journal* 165 255–74

**Weber E and Sonka S** 1994 Production and pricing decisions in cash-crop farming effects of decision traits and climate change expectations in **Jacobsen B, Pedersen D, Christensen J and Rasmussen S** eds *Farmers' decision making – a descriptive approach* European Association of Agricultural Economists, Copenhagen 203–18

**World Bank** 1998 *World Bank development report: knowledge for development* World Bank, Washington DC

**Ziervogel G** 2001 'Global science, local science: seasonal forecast use in a Basotho village' Open Meeting of the Global Environmental Change Research Community, 6–8 October 2001 Rio de Janeiro, Brazil

**Ziervogel G and Calder R** 2003 Climate variability and rural livelihoods: assessing the impact of seasonal climate forecasts *Area* 35 403–17

**Ziervogel G and Downing T E** forthcoming Stakeholder networks: improving seasonal forecasts *Climate Change*

14754959, 2004, 1, Downloaded from https://rgs-ibg.onlinelibrary.wiley.com/doi/10.1111/j.0016-7398.2004.05002.x by University Of Guelph, Wiley Online Library on [26/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License